

From Pseudo-Labels to Expert Grades: An Audit of DAB-Guided Weak Supervision and Conformal Calibration for HER2 IHC at Three Hospitals

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Abstract—HER2 immunohistochemistry (IHC) is scored 0/1+/2+/3+ from membrane-staining intensity, and no public dataset gives pixel-level annotations for it. We train a ResNet18-UNet on pseudo-labels from classical DAB optical-density thresholding and feed the network that same DAB channel as a fourth input. Against a SegFormer-B0 baseline under identical data, split and compute, a resolution-preserving decoder lifts moderate (2+) IoU from 0.00006 to 0.589 after four epochs, and to 0.657 (tissue mean IoU 0.811) after eight. We then audit the weak-supervision loop at three hospitals. Fed through the same three-feature head, the network’s area fractions grade the expert HER2 category as well as the threshold rule applied directly: accuracy 0.760 versus 0.750 on 200 held-out images (paired difference +0.010, 95% bootstrap interval [0.000, 0.025]), and no difference at BCI (0.299 versus 0.278) or ADEL (0.385 versus 0.373, $n = 418$). A network trained on pseudo-labels alone is therefore a learned approximation of its teacher. A multi-task model that adds expert image-level grades (and a stronger encoder and 38 times more patches) gains 15 points on the same in-domain images (0.900, paired +0.150, [0.075, 0.225]) but not significantly at a hospital it has never seen (BCI: +0.024 against a stain-normalised teacher; ADEL: +0.026, [−0.019, +0.072]), and the version trained on two sites collapses at the third (ADEL accuracy 0.196). Conformal prediction of the expert grade loses its guarantee at both other hospitals: standard split-conformal prediction promised 95% coverage and delivered 58.9% (BCI) and 74.2% (ADEL) for the model that never saw them; Macenko stain-similarity weighting raises BCI coverage to 63.6% and 91.7% for the two models, a partial repair; likelihood-ratio weighting reaches 99% only

by returning nearly every grade; 25 labelled local cases restore 95.9% (BCI) and 96.4% (ADEL), with sets of about 3.7 of 4 grades at ADEL. At the training hospital, splitting calibration and test sets by inferred slide instead of by patch leaves mean coverage nearly intact but widens its spread across calibrations several-fold. Targets are DAB pseudo-labels or public image-level grades, slide identity is inferred, and the other hospitals are single public datasets; we report what was measured and where it falls short.

Index Terms—HER2, immunohistochemistry, weak supervision, semantic segmentation, U-Net, conformal prediction, uncertainty quantification, digital pathology

I. INTRODUCTION

Human epidermal growth factor receptor 2 (HER2) status determines whether a breast cancer patient is eligible for HER2-targeted therapy. Pathologists assign a 0/1+/2+/3+ score from the intensity and completeness of membrane staining on an immunohistochemistry (IHC) slide, following the American Society of Clinical Oncology / College of American Pathologists (ASCO/CAP) guideline [1]. A score of 3+ is positive and 0/1+ is negative-to-low, but the 2+ (“equivocal”) category cannot be resolved from IHC alone and triggers a second, more expensive confirmatory test (FISH/ISH). Manual scoring, particularly at the 0-versus-1+ and 2+-versus-3+ boundaries, is known to vary between observers, which motivates quantitative, AI-assisted pre-scoring tools that reduce that variability

– not autonomous scoring, but a measurement a pathologist reviews and confirms.

Building such a tool for HER2 IHC segmentation runs directly into a data problem that most segmentation literature does not face: *no public dataset provides pixel-level HER2 IHC annotations*. The publicly available substitute we use, `HER2_IHC_40X`, provides only one intensity label per 1024×1024 patch and one overall score per source slide – exactly the classification-style labels a segmentation model cannot train on directly, and no slide identifiers at all, which rules out a true slide-level train/validation split. Any segmentation approach to this problem is therefore necessarily a *weakly-supervised* one, and the credibility of every downstream result depends on being explicit about what the model was actually trained and evaluated against.

A second gap is uncertainty. A pre-scoring tool that hands a pathologist a single point prediction per pixel, with no indication of when the model is unsure, invites over-trust at exactly the borderline cases where it matters most. Conformal prediction offers a distribution-free, finite-sample coverage guarantee for exactly this purpose, but existing work applying it to HER2 IHC under a real cross-institution stain shift is rarely examined, and conformal-prediction treatments of medical image segmentation [2], [3] are architecture- and task-agnostic uncertainty wrappers, not calibrated against a weakly-supervised, ordinal, clinically-defined class structure. We therefore evaluate conformal prediction where expert labels exist, at the level of the HER2 grade.

This paper makes four contributions:

- 1) A 4-channel (RGB + DAB optical density) input fusion for a ResNet18-UNet that resolves a diagnosed sub-pixel resolution failure of a transformer baseline on the class that decides reflex ISH testing (Section IV-E).
- 2) An *audit of the weak-supervision loop*: a test of whether the trained network carries information about the expert grade that its teacher, the DAB threshold rule, does not (it does not), and a comparison with a multi-task model that adds expert grades, which does in-domain but not significantly at an unseen hospital, and the version trained on two sites collapses at a third (Sections V-D and V-E).
- 3) An empirical study of conformal prediction of the expert HER2 grade under a real second-hospital stain shift: standard conformal loses its guarantee, Macenko stain-similarity weighting [10], [18] partially repairs it, and a few dozen local cases restore it. The same study shows that splitting by inferred slide rather than by patch widens the spread of coverage (Section V-F).
- 4) A formalization of the weak-supervision setting (Section IV-A) so that the circularity it introduces is a stated property of the estimator, and every result carries its caveats: pseudo-label targets, an inferred train/validation partition and a single second hospital. The segmentation network (ResNet18-UNet) and the image-level grade model used for conformal prediction (a ResNet-50 multi-task U-Net) are different models, and each section says

which it uses.

II. RELATED WORK

Computational pathology and HER2 IHC scoring automation. The broader shift toward AI-assisted diagnosis in digital pathology is well-documented [4], and HER2 IHC scoring specifically has attracted a growing body of automated work. Most of it targets image- or patch-level classification of the 0/1+/2+/3+ label directly – recent examples include pyramid-sampling convolutional architectures for whole-slide HER2 scoring [5] and multi-stage CNN classifiers for HER2 category prediction [6] – rather than pixel-level segmentation of the staining pattern itself, because pixel-level pathologist annotation is expensive and, for HER2 IHC specifically, not publicly available at scale. Our work targets the segmentation problem these patch-classification approaches sidestep, at the cost of having no direct pixel-level target to train on.

Weak supervision from classification labels. Where dense annotations are unavailable, a common strategy in histopathology is to derive a proxy dense target from a classifier-derived or classical, inspectable signal and train a segmentation model against it. Multi-layer pseudo-supervision from patch-level classification labels has been shown effective for tissue semantic segmentation more broadly [7], using class-activation-style pseudo-masks as the dense training target. We follow the same weak-supervision strategy, but our pseudo-labels are produced by a classical, physically interpretable rule (DAB optical-density thresholding via colour deconvolution [8]) rather than by a learned class-activation map, which keeps the pseudo-label generator itself auditable by a pathologist without requiring a second trained model to explain the first.

Stain variation and normalization. Colour and intensity variation across scanners and staining protocols is a well-known confound in computational pathology; deep generative approaches, including cycle-consistent adversarial networks for multi-domain stain normalization [9], address it by learning a normalizing transform. We take a complementary, non-generative approach: rather than transforming images into a canonical stain space, our stain-shift-weighted conformal predictor (Section IV-F) reweights calibration evidence by estimated stain similarity at inference time, using the same Macenko stain estimation [10] (Reinhard normalization [11] is also implemented but unused here, see Section IV-B).

Segmentation architectures. SegFormer [14] predicts at a coarse $H/4 \times W/4$ resolution before a final upsample, which is efficient for natural-image segmentation but, as Section IV-E shows, loses exactly the thin structures HER2 membrane staining is made of. U-Net [12], with skip connections carrying full-resolution encoder features into the decoder, is the architecture this failure mode motivates; we pair it with a pretrained ResNet18 [13] encoder. Focal loss [15] addresses per-pixel class difficulty, complementing Dice loss’s insensitivity to class cardinality on our severely imbalanced label distribution.

Conformal prediction. Inductive conformal prediction [16] produces set-valued predictions with a finite-sample marginal coverage guarantee, using a nonconformity score and an

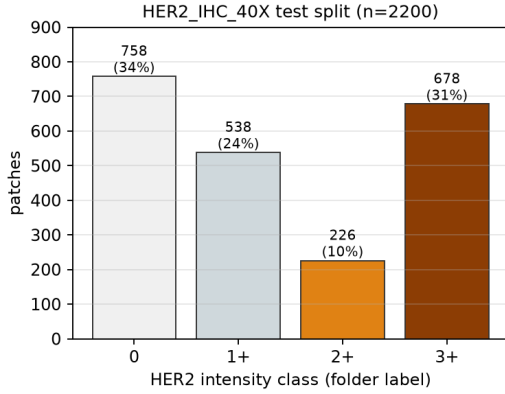


Fig. 1. HER2_IHC_40X test-split class distribution (folder label).

empirical calibration quantile. Sadinle et al. [17] give the multi-class treatment of the same construction. Tibshirani et al. [18] extend conformal prediction to covariate shift via importance-weighted calibration. Within medical imaging, conformal prediction has recently been used to wrap U-Net segmentation output with morphological prediction sets [2] and to quantify region-level segmentation uncertainty in clinical workflows [3]; both are general-purpose, task-agnostic wrappers over an already-trained segmentation model. We use the standard inductive construction for the four-grade image-level problem, add a stain-similarity weighting for cross-institution calibration, and measure what it does under a real second-hospital shift.

III. DATASET AND THE WEAK-SUPERVISION PROBLEM

We use HER2_IHC_40X, a public patch dataset of approximately 11,000 (10,997) 1024×1024 RGB IHC patches at 40× magnification, as an approved stand-in while a target clinical whole-slide-image dataset (84 slides, not yet digitized) remains unavailable. Patches are organised into four folders (`class_0`, `class_1+`, `class_2+`, `class_3+`); each patch’s filename additionally carries a prefix naming its source slide’s overall score. Cross-tabulating the two labels over all 10,997 patches produces an upper-triangular table – a patch’s folder-level label is never lower than its slide-level label – which establishes that the folder is a finer-grained, patch-level intensity label and the filename prefix is the coarser slide-level summary. Neither is a pixel mask. Fig. 1 shows the class distribution of the 2,200-patch test split used throughout this paper: moderately imbalanced, with the clinically pivotal moderate (2+) class the rarest at 10% of patches.

No slide identifiers. Filenames carry a sparse global counter, not a slide ID, so the coarsest available grouping is (slide-level score × origin token), 8 groups in total, and those groups are confounded with the label itself: holding out any one group removes an entire intensity class from training. We therefore cannot construct a true slide-level split. The held-out set used in this paper is instead defined by an origin token embedded in each filename, inferred to be the

TABLE I
SPLIT SIZES (SOURCE PATCHES; 40× NATIVE TILING PRODUCES 4 TILES PER PATCH). FIT/VALIDATION ARE OPTIMISTIC (NOT LEAKAGE-FREE); HOLDOUT IS THE BEST AVAILABLE APPROXIMATION TO A HELD-OUT SET ON THIS DATASET.

Split	Source patches	Tiles
Fit	200	792
Validation	30	119
Holdout (reserved)	1,904	–

TABLE II
PSEUDO-LABEL SANITY CHECK: MEAN DAB-THRESHOLDED STAINED-TISSUE AREA RISES MONOTONICALLY WITH THE DATASET’S OWN PATCH LABEL, AT TWO INDEPENDENTLY RETILED MAGNIFICATIONS.

Folder label	Stained area, 20×	Stained area, 40×
0	0.50%	0.49%
1+	3.84%	3.73%
2+	22.58%	21.47%
3+	63.57%	62.74%

dataset authors’ own train/test assignment – the dataset’s own `train/` and `test/` *directories* were measured to cross this token in both directions (78% of files in `test/` are named `_train_`), so using the directories directly would have held out a random resample of the same slides. Table I gives the resulting sizes. The validation set is a stratified random draw from the remaining patches and is explicitly *not* leakage-free; it is used only for epoch selection and curves, never quoted as a generalization estimate.

Pseudo-labels. Dense per-pixel targets are produced by thresholding DAB optical density (Section IV-B) at fixed cut points (weak 0.25, moderate 0.50, strong 0.80). We do not treat this as ground truth uncritically: mean stained tissue area rises monotonically with the dataset’s own patch-level label (Table II), at both 20× and 40× magnification, which is the expected signature of a sensible deconvolution and threshold set, and is the only sanity check available in the absence of real annotations.

Circularity. Because the segmentation model trains on these exact thresholds, a model that agrees with them has demonstrated nothing on its own. The classical thresholder is carried through every evaluation in this paper as an experimental control, not discarded once the learned model exists. Section IV-A states this circularity formally.

IV. METHOD

A. Problem Formulation and Notation

We make the weak-supervision setting explicit before describing the pipeline, since every later result inherits its properties from this setup. Let $x \in \mathbb{R}^{H \times W \times 3}$ denote an RGB patch and let $y^* : \Omega \rightarrow \{0, 1, 2, 3, 4\}$ denote the (unobserved) pixel-wise map a pathologist would assign over the patch’s spatial domain Ω , at the five intensity levels of Section III. No dataset available to this work observes y^* . Instead, a classical, deterministic thresholding rule $T_\tau : \mathbb{R}^{H \times W \times 3} \rightarrow \{0, \dots, 4\}^\Omega$, parameterized by fixed DAB optical-density cut points $\tau =$

(τ_1, τ_2, τ_3) (Section IV-B), produces a pseudo-label $\tilde{y} = T_\tau(x)$ used as a surrogate target.

We train a segmentation network $f_\theta : \mathbb{R}^{H \times W \times 4} \rightarrow \Delta^4$ – mapping a 4-channel (RGB + DAB) input to a per-pixel categorical distribution over 5 classes, Δ^4 the probability simplex – by minimizing an empirical risk with respect to $\tilde{y}$, not y^* :

$$\hat{\theta} = \arg \min_{\theta} \mathbb{E}_x \left[\mathcal{L}(f_\theta(x), T_\tau(x)) \right]. \quad (1)$$

Remark 1 (Circularity): Any statistic computed between $f_{\hat{\theta}}(x)$ and $\tilde{y} = T_\tau(x)$ – including every IoU, Dice, and conformal-coverage number in Section V – is, by construction of (1), a measurement of agreement with T_τ , not with y^* . It upper-bounds neither the model’s clinical accuracy nor its disagreement with a pathologist; it only tells us whether $f_{\hat{\theta}}$ has learned a spatially coherent, generalizing approximation of a known, inspectable rule. This is why the classical thresholder T_τ itself is retained as an experimental control throughout Section V: without it, an improving IoU against $\tilde{y}$ is uninterpretable, since it cannot be distinguished from the model simply memorizing T_τ more faithfully.

This formalizes, rather than merely asserts, why every empirical result below is reported as agreement with a rule, not as clinical validation – and it is the reason a genuinely useful next experiment (Section VI) is not a larger model or more epochs, but real pathologist annotations y^* to replace $\tilde{y}$ in (1) and in the conformal calibration of Section IV-F.

B. Preprocessing

HER2 IHC slides are dual-stained: haematoxylin (nuclear counterstain) and DAB (the HER2 signal). We separate the two with Ruifrok and Johnston colour deconvolution [8]: pixels are converted to optical density ($OD = -\log_{10}(I/I_0)$), and a 3×3 stain matrix (haematoxylin, DAB, and a residual direction from their cross product) is inverted to recover per-stain concentration maps. Fig. 2 shows this decomposition on a real, strongly-stained (3+) patch: the DAB channel isolates the membrane signal cleanly from the haematoxylin-stained nuclei, and is the quantity every intensity class and every area percentage in this paper is ultimately a function of. Stain normalization is implemented (Macenko [10], Reinhard [11]) but kept off for HER2_IHC_40X, which is single-source: normalizing would shift the exact DAB optical densities the classes are defined on, without correcting anything real.

Tissue is detected by mean optical density across RGB rather than by saturation-based Otsu thresholding: on this dataset, an Otsu threshold on saturation tracked staining intensity rather than tissue presence (the chosen threshold rose from ≈ 0.12 on faint patches to ≈ 0.43 on strongly stained ones), silently excluding weakly-stained tissue from its own denominator. Fig. 3 demonstrates this failure mode directly on a real faintly-stained patch: the saturation-Otsu mask covers only 34.9% of the frame, concentrated on the darkest haematoxylin-stained nuclei, while the optical-density mask covers 51.9%, recovering the faint, diffusely-stained tissue the

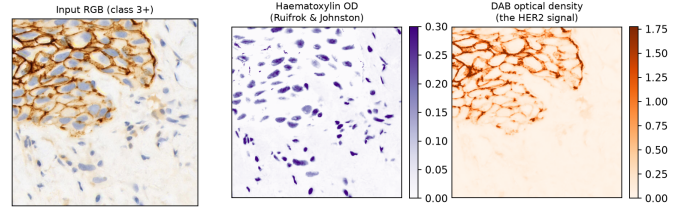


Fig. 2. Ruifrok-Johnston colour deconvolution on a real strongly-stained (3+) patch: input RGB, recovered haematoxylin channel, and recovered DAB channel (the HER2 signal every downstream measurement derives from).

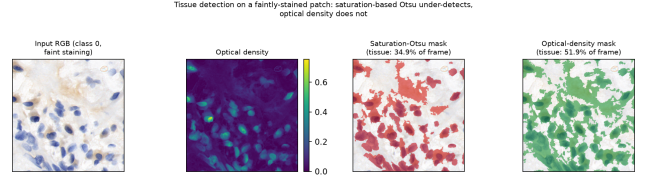


Fig. 3. Tissue detection on a real faintly-stained patch. Saturation-based Otsu thresholding under-detects tissue by discarding weakly-stained regions from its own denominator (34.9% vs. 51.9% of the frame); optical-density thresholding does not have this failure mode.

saturation-based rule discards. Optical density responds to any stain absorbing light, independent of which stain or how much.

Each 1024×1024 patch is tiled into four native-resolution 512×512 tiles (Section IV-E explains why native resolution matters), by strided subsampling rather than averaging, which would blend stained and unstained pixels and shift the optical densities the classes are defined on. Because the four tiles of one patch are near-duplicates of each other, the split (Table I) is drawn over source patches first and only then expanded to tiles, checked on every run to guarantee no patch has tiles on both sides of a split.

C. Architecture: ResNet18 Encoder, U-Net Decoder, 4-Channel Input

Our segmentation network pairs an ImageNet-pretrained ResNet18 encoder [13] with a U-Net decoder [12]. Skip connections carry the encoder’s own full-resolution early feature maps into the decoder, directly targeting the resolution failure diagnosed in Section IV-E, rather than relying on one coarse decode head and a single upsample to invent detail never actually predicted. A first version of the decoder ran one additional stage at the input’s own full resolution; measured cost (8 fit / 4 val patches, 1 epoch) was 170 seconds, four times the positions of the stage before it, so the decoder stops one stage earlier (stride 2, not stride 1) – still twice SegFormer’s stride-4 output resolution, at a cost this CPU-only machine can afford. Measured per-image cost (forward + backward, batch size 1, CPU) is ≈ 4.4 s, against SegFormer’s ≈ 1.45 s: about $3 \times$ slower, not an order of magnitude, for the resolution this architecture buys back (Fig. 7 quantifies this trade-off against the accuracy it purchases).

The network accepts four input channels, not three: RGB plus the DAB optical-density channel already computed in

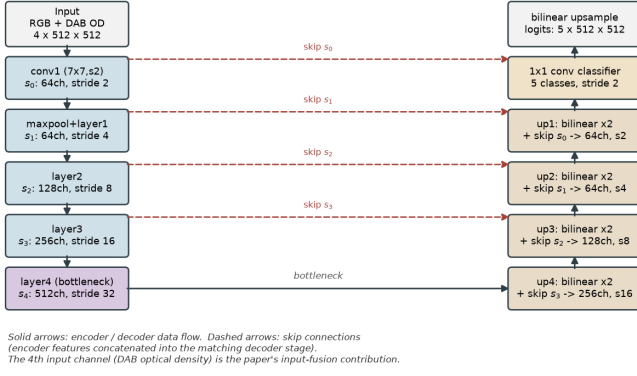


Fig. 4. ResNet18-UNet architecture, drawn from the model’s actual forward pass. The decoder stops at stride 2 (not stride 1), still twice SegFormer’s stride-4 output resolution, before a final non-learned bilinear upsample of the logits.

preprocessing. Because the pseudo-label targets are themselves a function of DAB optical density, this hands the model the exact, physically meaningful quantity its own weak-supervision targets are computed from, rather than requiring it to re-derive an approximation of DAB absorption from raw RGB through learned convolutions – a deliberate inductive bias for a weakly-supervised setting with very little signal for the rarest class. The first convolution’s extra input channel is initialized from the mean of the pretrained RGB filter weights, a standard transfer-learning trick for adding a channel without discarding ImageNet pretraining on the other three. Fig. 4 diagrams the full encoder-decoder, including exactly which stride the decoder stops at and why.

D. Loss

The class distribution is severely imbalanced at the pixel level (Fig. 5): background and negative jointly account for 80.8% of fit-set pixels, while the clinically pivotal moderate (2+) class is only 3.6%. We use a weighted sum of cross-entropy (weight 1.0), soft Dice (weight 0.5, Laplace-smoothed), and focal loss [15] (weight 1.0, $\gamma = 2$) to counteract this imbalance from three complementary angles: cross-entropy is the base likelihood objective; Dice is insensitive to class cardinality but sensitive to overlap quality; and focal loss downweights easy, well-classified pixels so gradient from the rare intensity classes is not drowned out by the two dominant classes. Dice is averaged only over classes actually present in a given batch – absent classes are excluded, not scored as 0 or 1, so a patch correctly containing no strong-staining pixels is not penalized for a class that was never there.

Inverse-frequency class weighting is implemented and can be enabled via a resolved "auto" config value (with a fail-out guard against silently training unweighted if resolution is skipped), but a full run comparing weighted against unweighted training was not completed within this paper’s compute budget; one completed epoch showed a moderate-class IoU of 0.0849 against 0.0001 at the equivalent unweighted

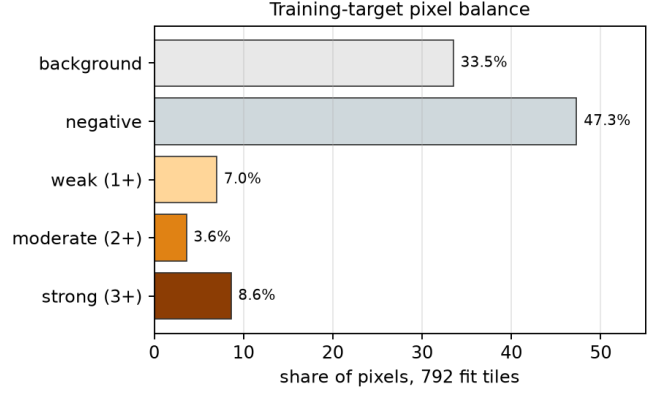


Fig. 5. Pixel-level class balance over the 792 fit-set tiles. Moderate (2+) is the rarest class by pixel count, not only by patch count (Fig. 1), which motivates the focal-loss and inverse-frequency-weighting treatments in Section IV-D.

point, which we report as preliminary evidence that class rarity is a contributing cause, not as a settled result (Section VI).

E. The Resolution Diagnosis

An initial SegFormer-B0 [14] run at effective $20\times$ (1024×1024 patches downsampled by 2 into one 512 tile) learned background, negative and strong staining, but essentially never learned the two middle classes (weak IoU 0.030, moderate IoU 0.000) despite a healthy overall pixel accuracy of 0.81 – a concrete demonstration of why pixel accuracy is the wrong metric to trust on this task. Qualitative inspection showed the mechanism: predictions were smooth blobs where the pseudo-label targets are thin membrane lace. SegFormer’s decode head predicts at $H/4 \times W/4$ and is bilinearly upsampled once at the end; at $20\times$ a membrane rim is 1–3 pixels wide, sub-pixel at the resolution the model actually predicts at, before the structure ever reaches the loss.

Retiling each 1024×1024 patch into four *native*-resolution 512×512 tiles, at identical model input size and compute cost, doubled every membrane’s width in pixels. Weak (1+) IoU improved $11\times$ ($0.030 \rightarrow 0.341$); tellingly, epoch 1 of the native-resolution run already beat the *entire* four-epoch downsampled run, on 40% of the data – the signature of a resolution problem, not a data-volume problem. **Moderate (2+) did not move:** IoU 0.0001, predicted on 253 of 21 million validation pixels. The confusion matrix showed the moderate row splitting 26% into weak and 71% into strong: the model treated the class as if it did not exist. This is the failure the U-Net decoder and the 4-channel input in this paper directly target.

F. Stain-Shift-Weighted Conformal Prediction

The method is defined for any softmax output $p(x) \in \mathbb{R}^C$: for pixels ($C = 5$) or for an image-level grade ($C = 4$). It follows the inductive conformal recipe with a multi-class score and stain-similarity weights. Section V-F evaluates it at image level, where expert grades exist; a pixel-level evaluation would be against pseudo-labels only and is not reported.

Nonconformity score. For a pixel with softmax output $p(x) \in \mathbb{R}^C$, the hinge nonconformity score for class c is

$$s_c(x) = 1 - p_c(x), \quad (2)$$

the multi-class least-ambiguous-set score [17].

Per-class calibration quantile. For each class c , calibration scores $\{s_c(x_i)\}_{i=1}^{n_c}$ are collected from every calibration pixel whose pseudo-label is c . The unweighted $(1 - \alpha)$ quantile is the standard inductive-conformal empirical quantile [16],

$$\hat{q}_c(\alpha) = \left\lceil \frac{(n_c + 1)(1 - \alpha)}{n_c} \right\rceil\text{-th smallest of } \{s_c(x_i)\}, \quad (3)$$

which gives the finite-sample marginal coverage guarantee. The stain-shift-weighted quantile [18] instead assigns each calibration score a similarity weight

$$w_i = \exp\left(-\frac{\|d_i - d_{\text{test}}\|^2}{2h^2}\right), \quad (4)$$

where d_i is a 6-dimensional stain descriptor (the haematoxylin and DAB rows of a per-patch Macenko-estimated stain matrix [10]) and h is the median pairwise descriptor distance over the calibration set (the standard "median trick" bandwidth). Weights are renormalized together with one reserved unit of weight for the test point itself – the Gaussian kernel evaluated at zero distance, so the test point's self-similarity weight sits on the same scale as every calibration weight rather than an arbitrary constant – and the weighted quantile is the smallest score at which cumulative normalized weight reaches $1 - \alpha$. Passing no test descriptor recovers the unweighted quantile exactly. The weights are a kernel-similarity heuristic, not an estimate of the likelihood ratio; we also evaluate the likelihood-ratio weights of [18] (a cross-fitted domain classifier on the same descriptors) for comparison.

Proposition 1 (Coverage under exchangeability): Under the standard inductive-conformal exchangeability assumption between the calibration pixels of class c and a test pixel truly of class c , the unweighted prediction set $\{c : s_c(x) \leq \hat{q}_c(\alpha)\}$ contains the true class with probability at least $1 - \alpha$, marginally over the randomness of the calibration draw [16]. The stain-shift-weighted variant restores this guarantee under a covariate shift between calibration and test stain distributions when the weights w_i correctly approximate the likelihood ratio between them [18]; Section V-F measures how far a heuristic kernel weight gets towards this under a real shift, and Section VI states why it cannot be expected to restore coverage fully: part of the shift is in how institutions grade, which is not a covariate shift at all.

Prediction sets and patch-level rollout. A pixel's prediction set is $\{c : s_c(x) \leq \hat{q}_c(\alpha)\}$. For a pixel-level use, each pixel's set reduces to a status – a definite class if the set is a singleton, AMBIGUOUS otherwise – and a patch's status is whichever definite class occupies a strict majority of its pixels, else AMBIGUOUS.

What this is calibrated against. No pathologist pixel annotations exist anywhere in this project. Calibration uses the same DAB-threshold pseudo-labels $\tilde{y}$ of Section IV-A. A

TABLE III
VALIDATION IOU BY CLASS, IDENTICAL DATA/SPLIT/SEED, 4 EPOCHS, CPU. VALIDATION IS A STRATIFIED RANDOM DRAW FROM THE FIT DIRECTORY, NOT A HELD-OUT SET (SEE TABLE I); NUMBERS ARE OPTIMISTIC AND USED FOR ARCHITECTURE COMPARISON, NOT AS A GENERALIZATION ESTIMATE.

Class	SegFormer-B0	ResNet18-UNet	Δ
Background	0.852	0.861	+0.009
Negative	0.815	0.839	+0.024
Weak (1+)	0.341	0.666	+0.325
Moderate (2+)	0.00006	0.589	+0.589
Strong (3+)	0.618	0.891	+0.273
Tissue mean IoU	0.443	0.746	+0.303
Pixel accuracy	0.862	0.908	+0.046

prediction set that achieves nominal coverage here has been shown well-calibrated against T_τ , not against y^* – this caveat travels with every conformal result in this paper, exactly as Remark 1 states for the segmentation metrics.

What is spent, once. Calibration and evaluation use only the reserved holdout set (Table I), split in half into a calibration set and a test set that never overlap and never touch the fit/validation data used for model selection.

V. EXPERIMENTS

A. Setup

All experiments run CPU-only (no GPU available), PyTorch, batch size 2 with gradient accumulation over 4 steps, AdamW at learning rate 10^{-4} with 5% linear warmup, seed 20260805. The architecture comparison (Section V-B) uses identical data, split and seed for both models, trained for 4 epochs, so the comparison isolates the architecture choice. The final U-Net (Table IV) was then trained for 8 epochs with the same settings and the epoch chosen on validation tissue mean IoU.

B. Architecture Comparison

Table III and Fig. 6 compare SegFormer-B0 and our ResNet18-UNet at full scale (200 fit / 30 validation patches, 4 epochs, identical split and seed). U-Net wins on every class, most dramatically on moderate (2+): IoU 0.00006 $\rightarrow$ 0.589, a result the smoke-scale run (40 fit / 15 val patches, 3 epochs) did not predict – at that scale moderate IoU *collapsed* across epochs (0.00022 $\rightarrow$ 0.00002), which we report because it initially supported the wrong hypothesis (that the failure was structural to the label distribution, independent of architecture). The full-scale run reverses that reading: epoch 1 with $5\times$ the data already showed non-trivial moderate-class signal (IoU 0.008), rising to 0.589 by epoch 4. The honest conclusion is that SegFormer's failure on this class is architecture-specific (consistent with the $H/4$ resolution diagnosis in Section IV-E), not an inherent property of the labels – but only the full-scale run, not the smoke-scale one, supports that conclusion, and we report both rather than only the one that agrees with our hypothesis.

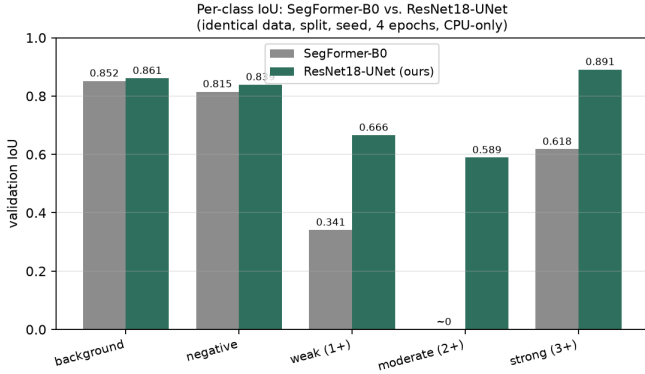


Fig. 6. Per-class validation IoU, SegFormer-B0 vs. ResNet18-UNet, same data/split/seed/epochs. Moderate (2+) – the class that decides reflex FISH testing – goes from effectively unlearned to the model’s third-best class.

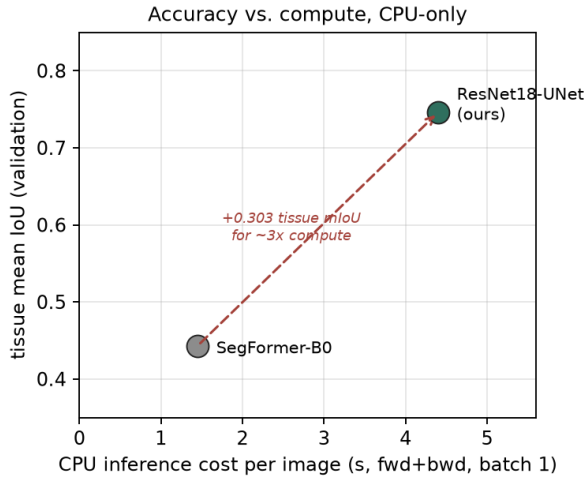


Fig. 7. The accuracy this architecture buys is not free, but it is cheap: +0.303 tissue mean IoU for approximately 3 $\times$, not an order of magnitude, more CPU inference cost per image.

C. Segmentation Results

Table IV gives full per-class metrics for the winning U-Net at its best epoch (4/4, selected on tissue mean IoU). Losses fell monotonically on both train and validation with no divergence (Fig. 8). Fig. 9 shows the row-normalized validation confusion matrix: moderate (2+) is still the most-confused class (12.9% of its pixels called weak, 11.5% called strong), but it is now a real, non-degenerate row rather than an empty one. Fig. 10 shows qualitative predictions against pseudo-label targets across all four staining intensities, with a per-example pixel-disagreement rate.

D. Audit: Does the Network Know More Than Its Teacher?

The network is trained to reproduce the threshold rule T_r (Section IV-A); agreement with T_r cannot show that it measures HER2. The non-circular test is downstream: take the per-image area fractions of weak, moderate and strong staining from (i) the network and (ii) the rule applied directly,

TABLE IV
PER-CLASS VALIDATION METRICS, RESNET18-UNET TRAINED FOR 8 EPOCHS, SELECTED EPOCH 6. SAME VALIDATION-SET CAVEAT AS TABLE III. THE 4-EPOCH MODEL OF TABLE III REACHED TISSUE MEAN IOU 0.746 AND MODERATE IOU 0.589.

Class	IoU	Dice	Precision	Recall
Background	0.896	0.945	0.940	0.950
Negative	0.898	0.946	0.954	0.938
Weak (1+)	0.778	0.875	0.849	0.903
Moderate (2+)	0.657	0.793	0.773	0.814
Strong (3+)	0.912	0.954	0.961	0.947
Tissue mean IoU: 0.811		Pixel accuracy: 0.938		

TABLE V
GRADING THE EXPERT HER2 CATEGORY FROM THE NETWORK’S VERSUS THE RULE’S AREA FRACTIONS (SAME HEAD). 95% BOOTSTRAP INTERVALS IN BRACKETS.

	Network	Rule	Difference
<i>Source holdout, n = 200</i>			
Accuracy	0.760 [.700, .815]	0.750 [.690, .805]	+ .010 [.000, .025]
QWK	0.904 [.869, .931]	0.900 [.865, .928]	
<i>BCI raw, n = 338</i>			
Accuracy	0.299 [.251, .346]	0.278 [.234, .328]	+ .021 [−.018, .059]
QWK	0.380 [.301, .450]	0.331 [.261, .396]	
<i>BCI site-normalised, n = 338</i>			
Accuracy	0.399 [.346, .453]	0.396 [.346, .447]	+ .003 [−.030, .033]
QWK	0.304 [.195, .397]	0.311 [.202, .401]	
<i>ADEL (third hospital, native colours), n = 418</i>			
Accuracy	0.385 [.340, .433]	0.373 [.328, .419]	+ .012 [−.012, .036]
QWK	0.322	0.529	

feed each through the same logistic head fitted on 200 source-site fit images, and compare each with the *expert* HER2 category. Evaluation: 200 held-out source images, 338 BCI images (second hospital) with raw colours and with site-level stain normalisation, and the 418 images of ADEL [20] (third hospital; centre-cropped to at most 1024 $\times$ 1024 at native scale, pixel size undocumented).

Table V shows no meaningful difference. On the source holdout the network is better by one point with an interval that touches zero, and at the second and third hospitals the differences are within noise (at ADEL the rule even has the higher weighted kappa, 0.529 against 0.322). The network therefore reproduces its teacher and inherits its failure at the second hospital; it does not improve grade-level information, so a network trained this way should be presented as a learned, fast approximation of an inspectable rule, and its value must rest on something the audit does not test (for example robustness to artefacts on which the rule fails), not on a grading gain. Both heads are fitted on one site and the audit is on 200 and 338 images, so small differences would not be detectable.

E. Adding Expert Supervision

The project’s second-generation model (“v2”) changes what the network learns from. It keeps a U-Net decoder trained on the same pseudo-label rule, and adds a score head on the encoder, trained on the real 0/1+/2+/3+ patch labels with attention pooling over the tiles of a patch. It also replaces the

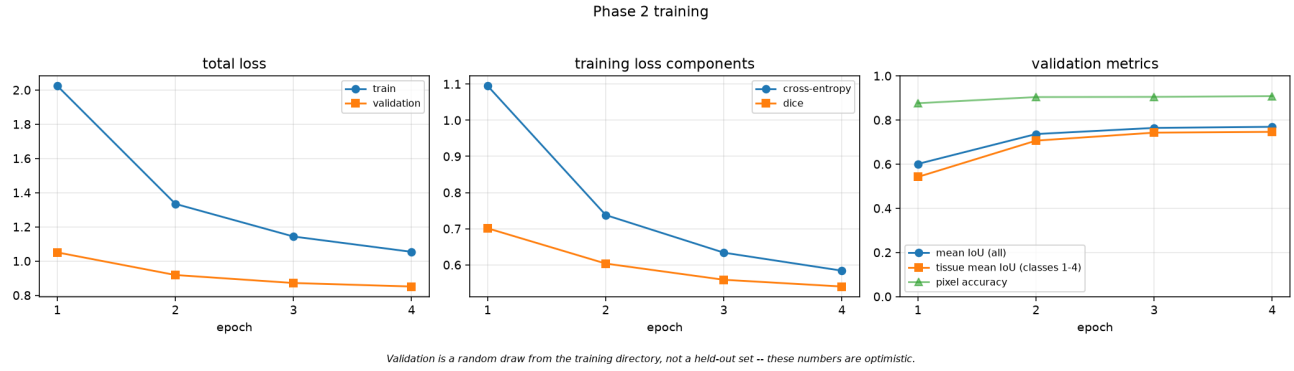


Fig. 8. Training loss, loss components, and validation metrics across the 4 epochs of the ResNet18-UNet run used in the architecture comparison. Validation numbers are optimistic (see Table I); shown for training diagnostics, not generalization claims.

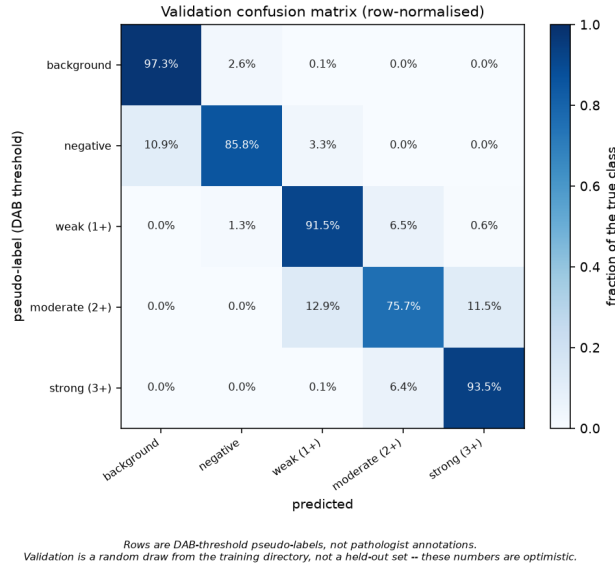


Fig. 9. Row-normalized validation confusion matrix. Rows are DAB-threshold pseudo-labels, not pathologist annotations.

TABLE VI

ACCURACY AGAINST THE EXPERT CATEGORY ON THE SAME IMAGES. HEADS IN THE FIRST TWO ROWS: DAB-PSEUDO-LABEL NETWORK AND THE THRESHOLD RULE (TABLE V). PAIRED DIFFERENCES ARE AGAINST THE RULE HEAD; 95% BOOTSTRAP INTERVALS. MODEL B HAS SEEN BCI'S TRAINING SPLIT, SO ITS BCI COLUMN IS A SEEN-SITE RESULT; ADEL IS UNSEEN BY BOTH.

	Source (200)	BCI raw (338)	vs norm. rule	ADEL (418)
Pseudo-label net + head	0.760	0.299	–	0.385
Threshold rule + head	0.750	0.278	0.396 (norm.)	0.373
v2 Model A	0.900 [.86, .94]	0.420 [.37, .47]	+ .024 [–.03, .08]	0.400 [.35, .45]
paired vs rule (raw)	+ .150 [.08, .23]	+ .142 [.09, .20]		+ .026 [–.02, .07]
v2 Model B (BCI seen)	0.915 [.88, .95]	0.722 [.68, .77]	–	0.196 [.16, .23]

ResNet18 encoder with a ResNet-50 initialised from pathology self-supervised weights and trains on all 7,729 fit patches instead of the 200 that fit a CPU budget, with stain, zoom, blur and noise augmentation. *Model A* trains on the training site only; *Model B* adds BCI's training split. Table VI compares them with the audit's two heads on the *same* images.

In the training domain the expert-supervised model is 15

points better than the teacher-derived features, with a paired interval far from zero (Model A 0.900 versus 0.750; QWK 0.953 versus 0.900). At a hospital it never saw (Model A on BCI, raw colours) it is 14 points better than the raw-colour teacher, but the teacher with site-level stain normalisation (0.396) closes almost all of that (difference +0.024, interval [–0.033, 0.080], not significant): the cross-site gain is mostly a colour correction available to the teacher too. Model B reaches 0.722 on BCI only because it trained on that hospital, and at ADEL, which neither model has seen, the two-site model B predicts almost every image 2+ (accuracy 0.196, QWK 0.04, AUC for 0 versus 1+ 0.53; 20.3 points below Model A, paired interval [–26.6, –14.3]), while Model A is 2.6 points above the rule, not significantly. Training on more sites did not make the model robust to a further hospital; it made it worse. Two limits apply. The v2 model differs from the audited network in three ways at once (expert labels, 38 times more data, a stronger encoder), so the table shows what the full system gains, not which factor does it; separating them needs a pseudo-label-only run at v2 scale, which our budget did not allow. And the 200-image subset is balanced by grade, so accuracies are not those of the full 1,904-image holdout (Model B: 92.3%, QWK 0.975).

F. Conformal Prediction Under a Second Hospital

Setup. The unit is an image-level HER2 grade (0/1+/2+/3+) with the expert grade as label and the least-ambiguous-set score. We calibrate on 951 held-out training-site images and test on the other 953 (same hospital) and on 338 test images of BCI [21] (a second hospital, different scanner; case scores from the pathology report). *Run A* is a model that never saw BCI, a genuine cross-institution test; *Run B* is the deployed model, which was trained with BCI's training split. The kernel stain weights use the corrected Macenko descriptor (Section IV-B); an earlier version of this study used a descriptor with an orientation fault and is superseded. Target coverage is 95%.

Results. At the training hospital every method holds its target (Run B: 93.6–94.9% at a 95% target, 89.1–90.0% at 90%). At the second hospital (Table VII) standard conformal

Phase 2 predictions vs DAB-threshold targets (validation patches)

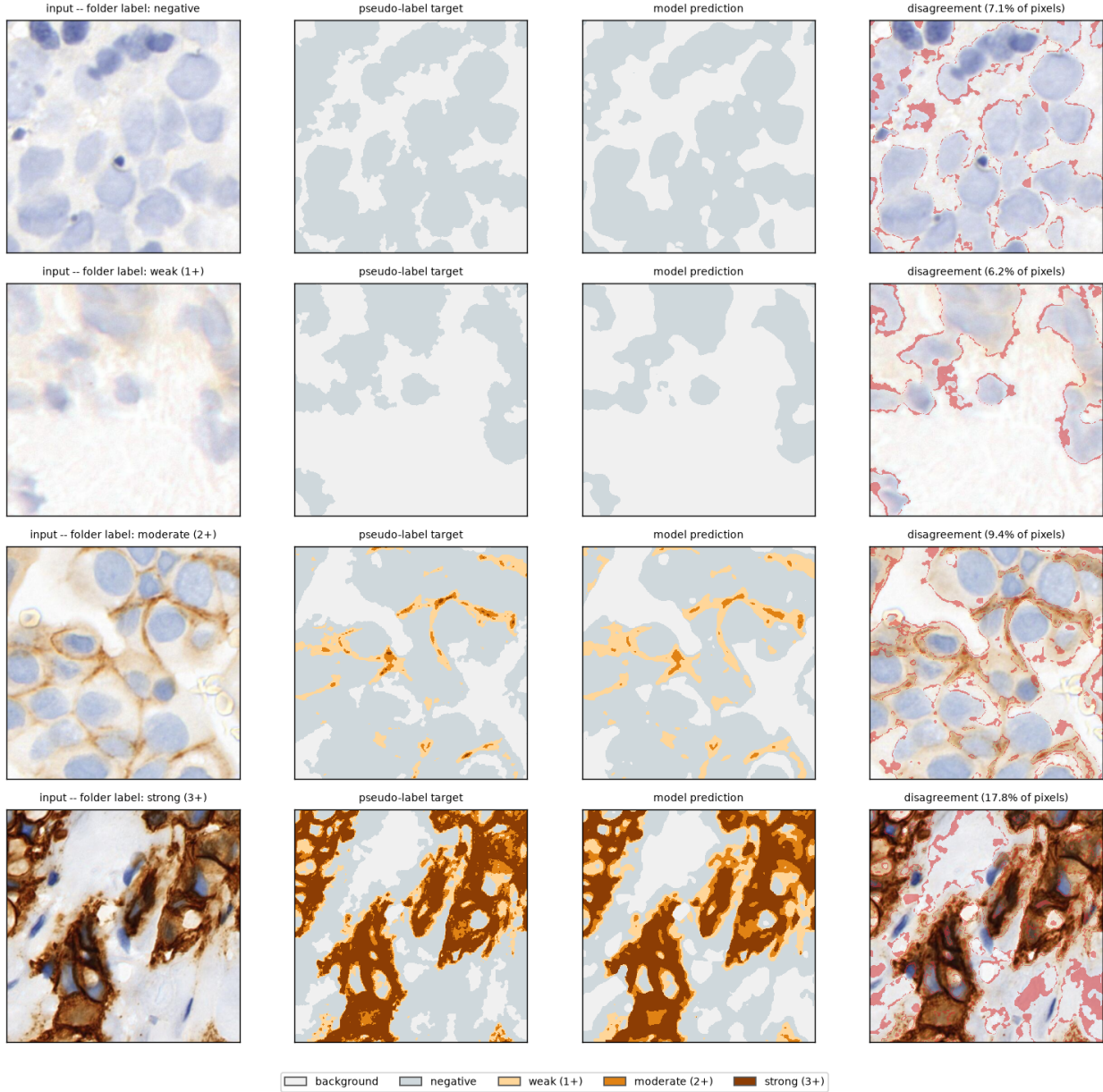


Fig. 10. Input, pseudo-label target, model prediction and pixel disagreement for one validation patch per intensity class. The model recovers thin membrane structure the pseudo-label targets are made of, rather than the smooth-blob failure diagnosed in Section IV-E.

prediction loses its guarantee: 95% promised, 58.9% (Run A) and 82.2% (Run B) delivered. Stain-similarity weighting improves both, by 4.7 and 9.5 points, and is the method's real, measured contribution; it does *not* restore coverage, because part of the shift is in how the two institutions grade, which no covariate weighting can correct. Likelihood-ratio weighting restores validity but only by returning nearly every grade: the two hospitals' stain features are almost perfectly separable (domain AUC 0.999, effective calibration size 2), so the weights concentrate on a handful of cases and the sets become

uninformative. At ADEL, a hospital whose images neither model has seen, unweighted conformal prediction delivers 74.2% for Model A and 28.0% for Model B at a 95% target, with the oracle-prior label-shift correction at 78.7% and 38.8%, and 25 local labelled cases restore 96.4% and 96.3%, with sets of 3.7 and 3.9 of 4 grades, which is close to returning every grade. A label-shift correction [19], which reweights calibration cases by the ratio of target to source class priors, is the principled alternative to the kernel heuristic when the shift is in case mix (BCI has far fewer grade-0 cases

TABLE VII
COVERAGE AT THE SECOND HOSPITAL, TARGET 95%.

Calibration	BCI		ADEL	
	Run A	Run B	Run A	Run B
Unweighted	58.9	82.2	74.2 (2.4)	28.0 (1.3)
Stain-similarity weighted (kernel)	63.6	91.7	—	—
Label-shift weighted, oracle prior [19]	62.7 (1.7)	92.6 (1.7)	78.7 (2.6)	38.8 (1.7)
Label-shift weighted, estimated prior	54.1 (1.3)	78.7 (1.1)	—	—
Likelihood-ratio weighted	99.7 (3.9)	99.1 (3.7)	—	—
25 local labelled cases	95.9 (3.8)	95.9 (2.1)	96.4 (3.7)	96.3 (3.9)

Coverage in %, target 95%; mean set size (of 4 grades) in parentheses where measured. Stain-similarity weights need stain descriptors, which were extracted for BCI only. BCI: 338-image test subset; ADEL: all 418 images.

TABLE VIII
SAME-HOSPITAL COVERAGE OF THE LAC CONFORMAL SET, PATCH SPLIT VERSUS PSEUDO-SLIDE SPLIT (300 CALIBRATIONS EACH). MEAN, STANDARD DEVIATION AND 5TH PERCENTILE OF COVERAGE ON THE UNSEEN HALF.

Model	Target	Split	Mean	SD	5th pct.
B	95%	patch	0.950	0.010	0.934
B	95%	slide	0.939	0.038	0.878
B	90%	patch	0.901	0.014	0.879
B	90%	slide	0.877	0.080	0.711
A	95%	patch	0.950	0.010	0.931
A	95%	slide	0.924	0.066	0.775
A	90%	patch	0.901	0.013	0.880
A	90%	slide	0.871	0.107	0.648

than the training site). With the oracle target prior (the true class proportions of the BCI test images, which a laboratory would not know) it reaches 92.6% for the deployed model, a point above the stain weights, and only 62.7% for the model that never saw BCI; with the prior estimated from unlabelled predictions by confusion-matrix inversion it is worse than not weighting at all (78.7%, 54.1%). Case mix is therefore part of the shift but not all of it, and a prior estimated from an inaccurate model cannot be trusted. Calibrating on 25 labelled cases from the new hospital restores 95.9% coverage for both models and, for the deployed model, with sets of 2.1 grades on average. The practical reading is that cross-institution conformal calibration needs local labelled cases, and that stain weighting buys a smaller but real improvement before they are available.

Which model. The conformal study uses the image-level grade model (Run A/Run B, a ResNet-50 multi-task U-Net with a grade head), not the ResNet18-UNet of Section V-B; the stain-similarity weights use the Macenko descriptor either way.

Slide split versus patch split. HER2_IHC_40X has no slide identifiers, so a random split of patches shares slides between calibration and test sets. Patch indices are consecutive within a patch run and the patch grade is constant along each run, so we form 51 *pseudo-slides* (maximal runs with the same prefix and grade and no index gap above 20; a proxy that can merge slides but not split a run), and compare 300 calibrations by patch with 300 by whole pseudo-slides at the training hospital (Table VIII).

Mean coverage is nearly intact (0.94 at a 95% target for model B, 0.88 at 90%), as marginal conformal theory allows,

but its spread across calibrations grows four- to eight-fold and its 5th percentile falls to 0.65–0.88: a laboratory calibrating on a handful of slides cannot count on the average. The calibration-hospital numbers above therefore describe an average over many random patch draws, which patch-level splits make look more stable than slide-level deployment would be.

Not shown. One second hospital, one scanner and case-level rather than field-level labels; no repeated model training; and the weighting is a heuristic without a coverage guarantee under concept shift.

VI. THREATS TO VALIDITY

We organize this section by the standard empirical-software-engineering taxonomy of validity threats, since a weakly-supervised medical imaging study has representatives of every category and conflating them obscures which threats future work can and cannot address by better engineering alone.

Construct validity. Every reported IoU, Dice, precision, recall and conformal-coverage number is a measurement of agreement with the classical DAB-threshold rule T_τ (Remark 1), not with pathologist judgment y^* . This is not a caveat that shrinks with more data or longer training – it is definitional to (1), and only real pixel-level pathologist annotation resolves it. Relatedly, ASCO/CAP scoring [1] is defined on percentage of tumour cells; every measurement here is percentage of tissue area, which includes stroma, and the two diverge whenever stroma content varies (mean tissue fraction 0.45 in class 0 to 0.79 in class 3+).

Internal validity. The held-out set in Table I contains all 1,847 patches of the dataset authors’ WSI-based test split (slides disjoint from their training slides) plus 57 further patches, so it is slide-disjoint from the authors’ training slides. It is not guaranteed slide-disjoint from our fitting data: 1,000 of the 9,093 fit and validation patches (11%) are outside the authors’ WSI-based training set and their slides are unknown. The validation set used for epoch selection is a random draw of patches from the training directory and is optimistic. HER2_IHC_40X carries no slide identifiers, so we cannot verify either statement, and the authors’ test split holds only about 21 whole slides. Section V-F quantifies the effect for conformal coverage with 51 inferred pseudo-slides: the mean is nearly unchanged but the spread across calibrations grows several-fold.

External validity. Three public datasets, two external hospitals, each a single source. The conformal study shows that coverage fails at both and partly recovers at one; ADEL images were centre-cropped and padded without a documented pixel size, so part of the drop there may be a scale mismatch. BCI labels are case-level scores from a report and BCI images are cropped to 512 px, so a crop of a 3+ case may contain no stained tumour, and the labels differ in kind from the training site’s patch grades. No claim here is evidence of performance on the clinical population this project ultimately targets, whose slides are not yet digitized.

Statistical conclusion validity. The conformal coverages are computed on 951 calibration and 953 + 338 test images,

with the local-calibration row averaged over 50 random splits; the other rows are single evaluations. The architecture comparison and the segmentation results are single-seed (CPU-only compute budget) and should be read as directional. The inverse-frequency class-weighting result in Section IV-D is a single completed epoch, reported as preliminary evidence.

The remaining open question. Moderate (2+) is resolved as *learnable* (Table III), not as *solved*: at IoU 0.657 (8 epochs; 0.589 at 4) against 0.78–0.91 for the other stained classes, it remains the weakest class by a real margin, and it is precisely the category that decides reflex FISH testing. We conjecture, though do not yet test, that the highest-value next step is not a further architecture search but reframing 2+ as a per-cell membrane-completeness question rather than a tissue-area intensity question – the approach ASCO/CAP itself scores by, which our pixel-area measurement only approximates.

VII. BROADER IMPACT AND ETHICAL CONSIDERATIONS

We designed this system around one non-negotiable constraint: it *measures*, and a pathologist *scores*. Every quantity this paper reports is an area percentage or a calibrated prediction set, never a categorical HER2 call, and the surrounding system this method feeds into enforces that boundary as executable code rather than as a documentation promise – an engineering decision motivated directly by the automation-bias risk that any sufficiently accurate-looking assistive tool for a high-stakes clinical decision invites over-reliance. Conformal prediction’s role in that design is specifically to make the model’s uncertainty visible rather than to hide it behind a single confident number; a wide prediction set at a clinically relevant α is a legitimate, intended output, not a failure mode to be tuned away.

Two further considerations bear directly on responsible deployment. First, representativeness: HER2_IHC_40X’s demographic and institutional provenance is not documented at the level clinical deployment would require, and every result in this paper should be read as evidence about this dataset, not as evidence of population-level clinical validity – Section V-F shows that even a modest second-hospital shift breaks nominal coverage, and that local calibration, not stain weighting alone, is what restores it. Second, equity of access: by design, every component in this pipeline runs on CPU-only commodity hardware; a resource-constrained pathology department without GPU infrastructure is not, on that basis alone, excluded from using tooling of this kind. We view this as a deliberate, not incidental, property of the engineering choices in Section IV-E and Section IV-D.

VIII. CONCLUSION AND FUTURE WORK

We show that a physically-grounded input channel (DAB optical density, computed by the same deconvolution the weak-supervision targets derive from) combined with a resolution-preserving decoder resolves a specific, diagnosed failure of a transformer segmentation baseline on HER2 IHC’s most clinically consequential class, under a strict CPU-only compute

budget. We further show, on a real second hospital, that standard conformal prediction of HER2 grade loses its coverage guarantee (95% promised; 58.9% and 82.2% delivered), that stain-similarity weighting repairs part of the loss but not all of it (63.6%, 91.7%), and that a few dozen labelled local cases restore it (95.9%). We formalized the weak-supervision setting this all sits inside (Section IV-A) precisely because, on a problem with no verified slide-level split and no pixel-level ground truth, a number reported without the estimator properties that produced it is not reproducible, however precise it looks.

Future work follows directly from Section VI: further institutions, to test whether stain weighting’s partial repair is stable; multi-seed replication of the architecture comparison; real pathologist pixel annotations to replace pseudo-label supervision; and a per-cell membrane-completeness reformulation of the moderate (2+) class, closer to how ASCO/CAP itself defines the score.

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